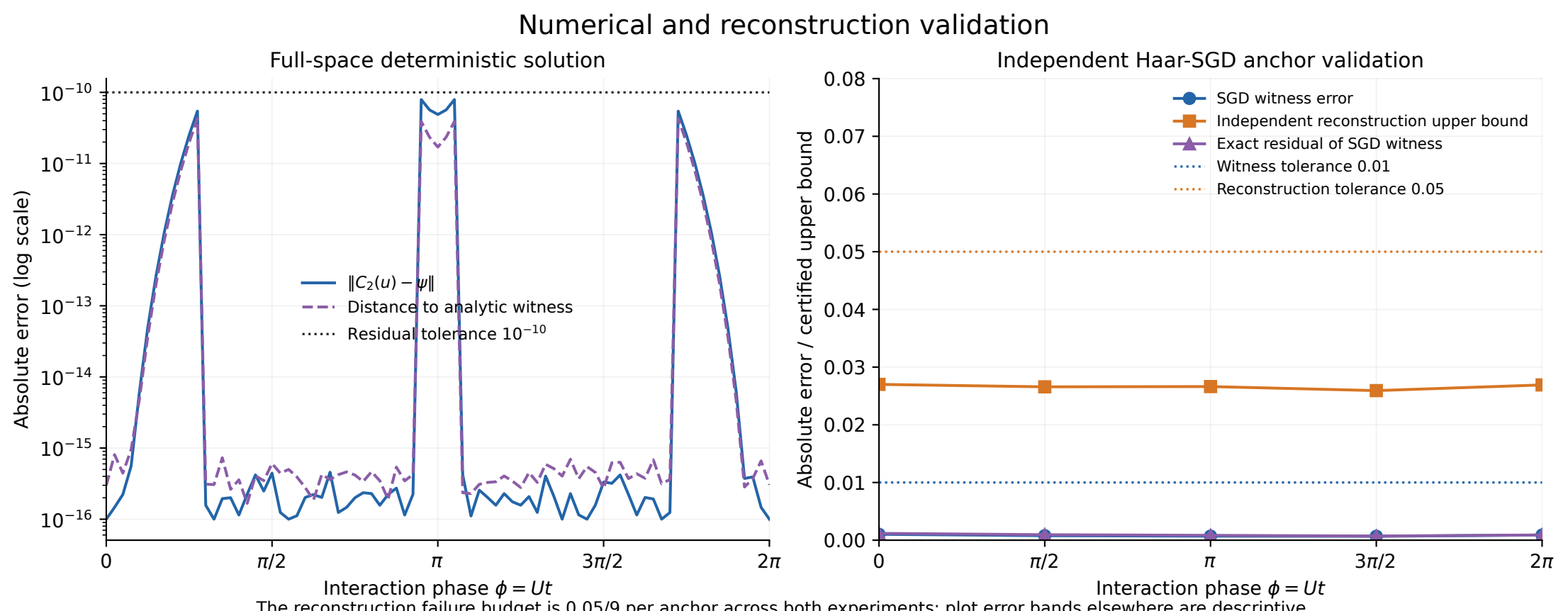
Numerical and reconstruction validation
Full-space deterministic solution
Absolute error (log scale)
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
10^{-15}
10^{-16}
$\|C_2(u) - \psi\|$
Distance to analytic witness
Residual tolerance 10^{-10}
0
π/2
π
3π/2
2π
Interaction phase $\phi = Ut$
Independent Haar-SGD anchor validation
Absolute error / certified upper bound
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
SGD witness error
Independent reconstruction upper bound
Exact residual of SGD witness
Witness tolerance 0.01
Reconstruction tolerance 0.05
0
π/2
π
3π/2
2π
Interaction phase $\phi = Ut$
The reconstruction failure budget is 0.05/9 per anchor across both experiments; plot error bands elsewhere are descriptive.
Errors below 1e-16 are displayed at 1e-16 on the logarithmic axis.